A Text-guided Protein Design Framework

April 15, 2026

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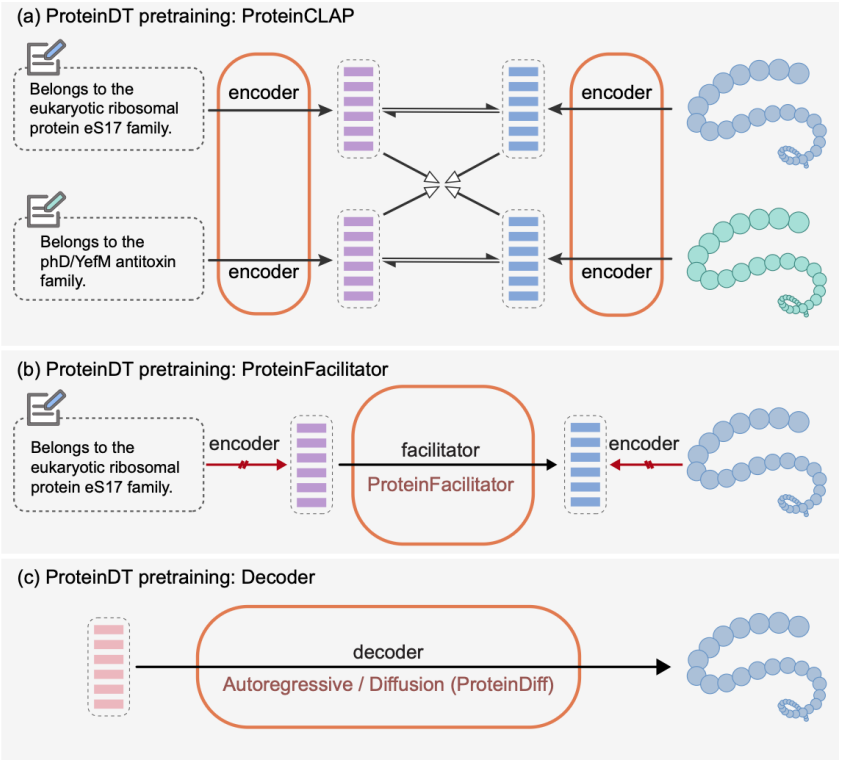
[1] Liu, Shengchao, et al. "A text-guided protein design framework." Nature Machine Intelligence 7.4 (2025): 580-591.

Target

Use **text descriptions** to design **protein**.

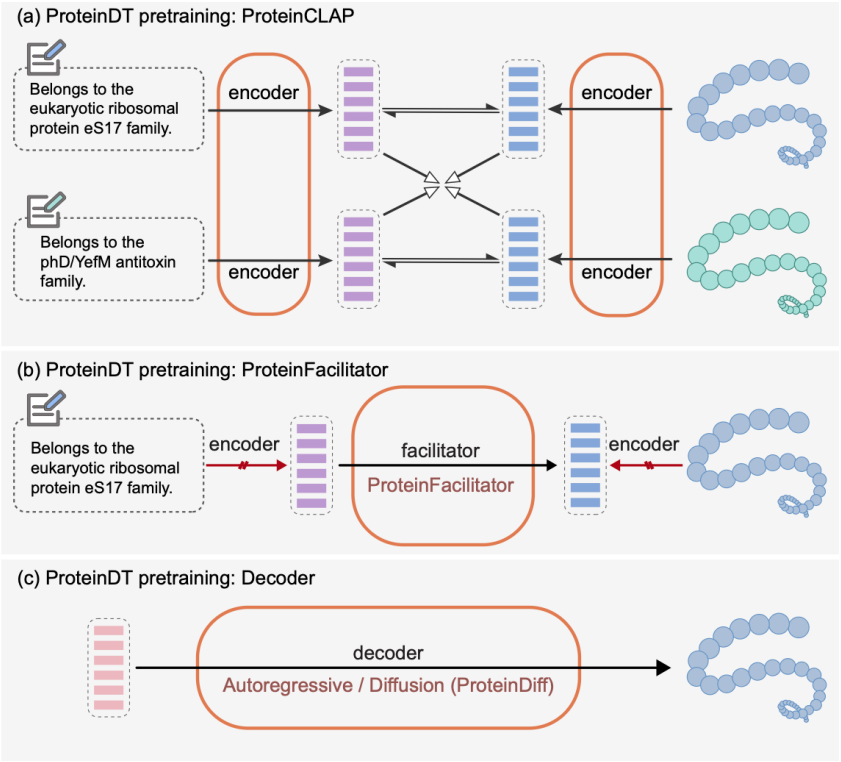
- text-to-protein generation,
- zero-shot text-guided protein editing,
- protein property prediction.

Model-pretraining



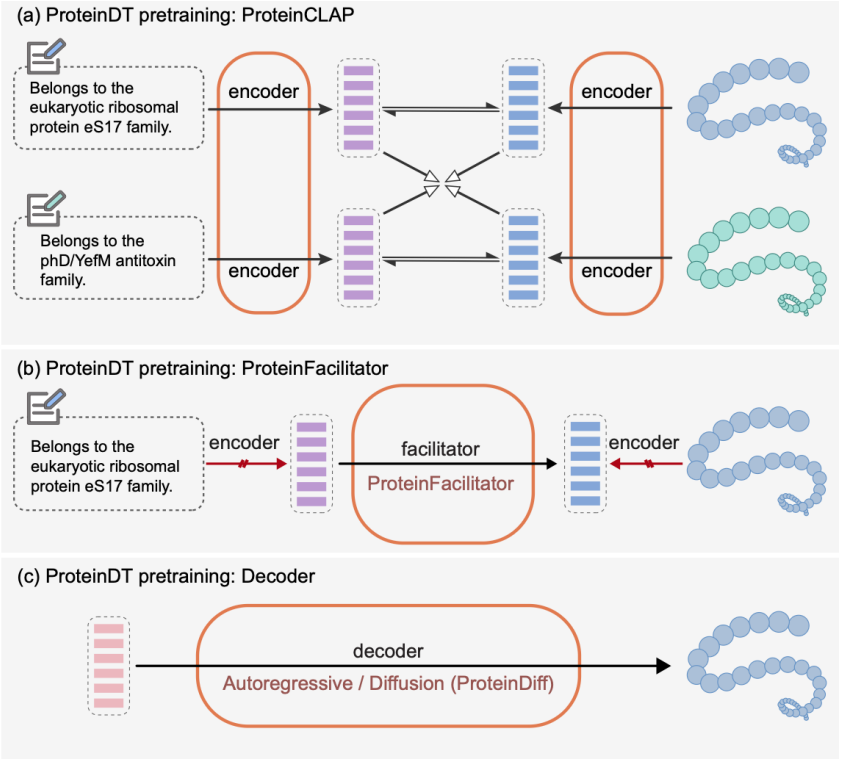
(a) **Representation alignment:** BERT-like models for encoder. Use contrastive learning to encode text sequence x_t and protein sequence x_p into representations z_t and z_p.

Model-pretraining



(b) **ProteinFacilitator:** map the text representation z_t to a protein-like representation z_p'. Introduce an alignment module $\mathrm{ProteinFacilitator}()$ and then we estimate the it using a conditional Gaussian as $p(z_p|z_t) = N(z_p; \mathrm{ProteinFacilitator}(z_t), I)$.

Model-pretraining



(c) **Protein decoding:** use the predicted protein representation to generate the corresponding protein sequence:

$$x_t \rightarrow z_t \rightarrow z_p^{\prime} \rightarrow x_p.$$

Downstream Task: Text-to-Protein Generation

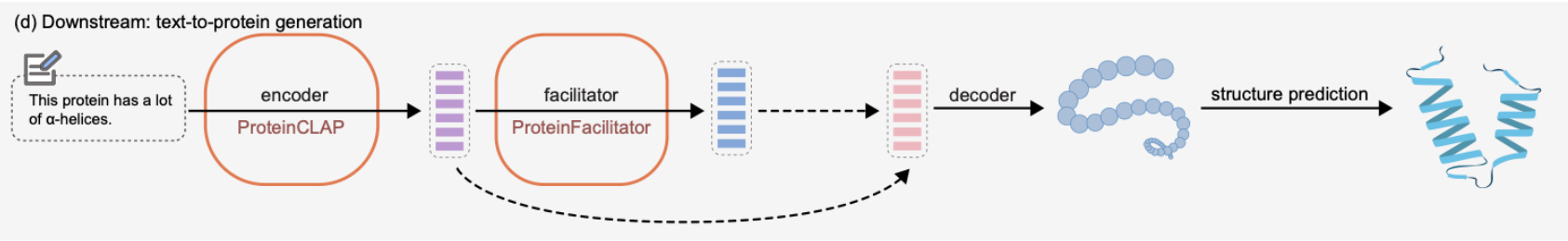
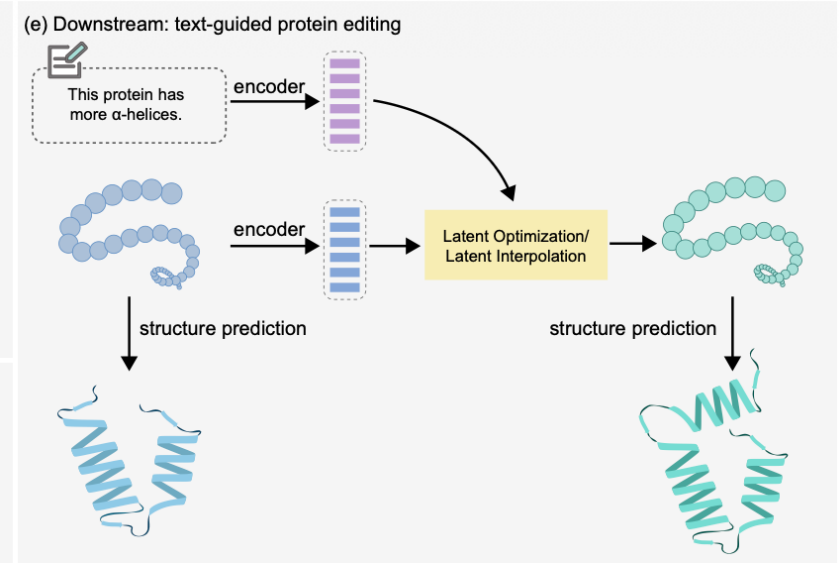


Table 1. Retrieval accuracy (%) for text-to-protein generation. It is a multiple-choice task (T options), measuring the consistency between the input text prompt and the generated protein sequence.

	Galactica	ChatGPT	ProteinDT (Ours)					
			w/o ProteinFacilitator			w/ ProteinFacilitator		
	AR	AR	AR	ProteinDiff		AR	ProteinDiff	
			Transformer	RNN	Transformer	Transformer	RNN	Transformer
T = 4	51.5	38.5	49.00	24.00	35.50	97.00	40.50	51.50
T = 10	29.0	23.0	27.00	10.50	17.50	91.00	21.50	25.00
T = 20	19.0	15.5	20.00	5.50	9.50	83.50	15.00	13.50

Galactica and ChatGPT. It is noteworthy that both baselines are Transformer-based autoregressive models.

Downstream Task: text-guided protein editing



Latent Interpolation: Apply the spherical linear interpolation (SLERP) [13], which yields an intermediate representation $\tilde{z}^c = \mathrm{SLERP}(z^t, z^p, \theta)$, and the output protein is generated from decoder, as $\tilde{x}^p \sim p(x^p \mid \tilde{z}^c)$.

Latent Optimization: It learns a token-level latent code z^w that is close to both the text and protein sequence-level representation, as

$$z^w = \arg\min \left(\lambda \|P(z^w) - z^t\|^2 + (1-\lambda)\|P(z^w) - z^p\|^2\right)$$

Downstream Task: text-guided protein editing

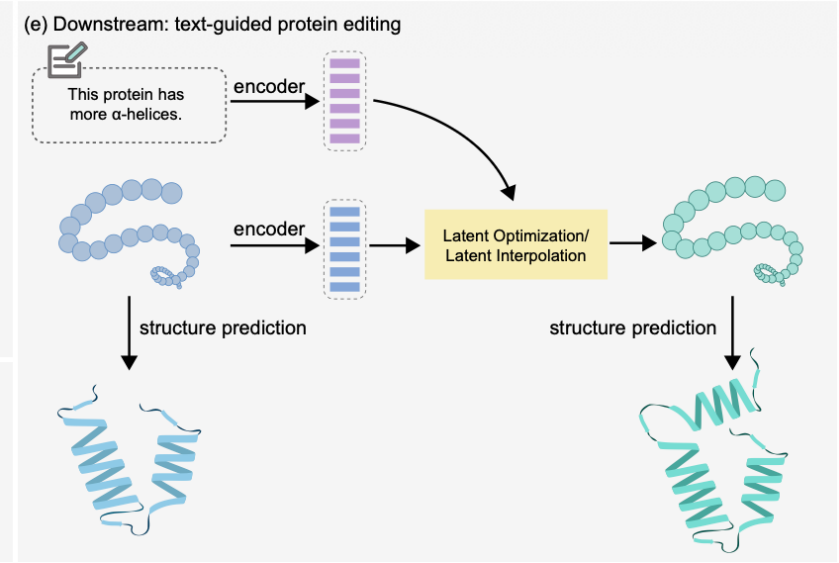


Table 2. The satisfactory hit ratio (%) of text-guided protein editing tasks. The **best** results and the **second-best** results are marked.

			Structure: α-helices				Structure: β-sheets				Region	
			more		less		more		less		ordered	disordered
			BERT	DSSP	BERT	DSSP	BERT	DSSP	BERT	DSSP	pLDDT	pLDDT
Galactica			5.46	7.02	13.26	10.53	1.95	2.14	11.50	11.31	**8.47**	23.73
ChatGPT			5.46	6.82	7.80	6.43	4.68	2.92	5.07	3.51	6.78	62.71
ProteinDT (Ours)	Latent Interpolation	ProteinDiff-RNN	**67.64**	**75.63**	19.49	10.14	27.88	11.31	**53.61**	**69.98**	5.08	**94.92**
		ProteinDiff-BERT	38.79	51.27	**48.15**	**34.70**	**39.38**	**15.79**	42.30	65.50	3.39	**94.92**
		AR-BERT	27.49	45.61	**56.92**	**39.38**	9.16	5.26	**69.79**	**73.10**	**15.25**	84.75
	Latent Optimization		**51.66**	**54.19**	33.92	32.94	**46.59**	**39.57**	39.38	46.59	6.78	91.53

			Stability: Villin				Stability: Pin1				binding editing			
			more stable		less stable		more stable		less stable		higher		lower	
			BERT	Rosetta	BERT	Rosetta	BERT	Rosetta	BERT	Rosetta	BindingNet	GNINA	BindingNet	GNINA
Galactica			7.81	51.56	73.44	31.25	1.41	4.23	8.45	4.23	26.61	21.77	33.87	25.00
ChatGPT			**29.69**	**57.81**	78.12	28.12	**28.17**	19.72	39.44	71.83	16.94	12.90	26.61	35.48
ProteinDT (Ours)	Latent Interpolation	ProteinDiff-RNN	17.19	50.00	82.81	54.69	4.23	**26.76**	95.77	70.42	**58.87**	**29.84**	37.90	**70.16**
		ProteinDiff-BERT	9.38	23.44	**93.75**	**68.75**	2.82	16.90	**97.18**	**84.51**	50.81	29.03	43.55	67.74
		AR-BERT	6.25	40.62	**93.75**	**59.38**	1.41	25.35	**98.59**	**73.24**	**53.23**	**29.84**	**45.97**	**71.77**
	Latent Optimization		**50.00**	**53.12**	54.69	40.62	**52.11**	**35.21**	47.89	63.38	39.52	**37.90**	**60.48**	61.29

Downstream Task: Protein Property Prediction

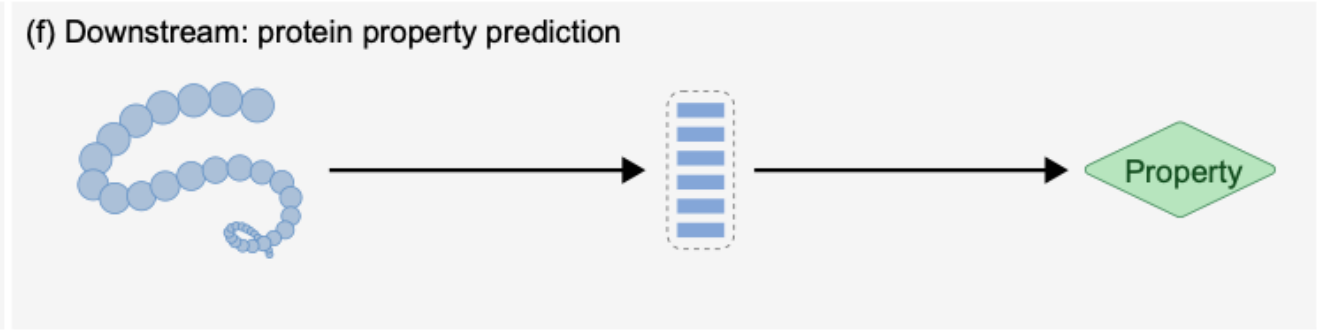


Table 3. Results on six protein property prediction tasks from TAPE. The **best** results and the **second-best** results are marked. Baseline results are replicated from OntoProtein, with the exception of ProtBERT and OntoProtein for Contact prediction, both of which were rerun.

Method	Structure			Evolutionary	Engineering	
	SS-Q3 ↑	SS-Q8 ↑	Contact ↑	Homology ↑	Fluorescence ↑	Stability ↑
LSTM	0.75	0.59	0.26	0.26	**0.67**	0.69
TAPE Transformer	0.73	0.59	0.25	0.21	**0.68**	0.73
ResNet	0.75	0.58	0.25	0.17	0.21	0.73
MSA Transformer	-	0.73	0.49	-	-	
ProtBERT	0.81	0.67	0.59	**0.29**	0.61	**0.82**
OntoProtein	0.82	0.68	0.56	0.24	0.66	0.75
ProteinDT-ProteinCLAP-InfoNCE (Ours)	**0.8331**	**0.6935**	**0.6031**	**0.3092**	0.6029	**0.8130**
ProteinDT-ProteinCLAP-EBM-NCE (Ours)	**0.8334**	**0.6919**	**0.6007**	0.2841	0.6105	0.7952